Building Blocks

Module 13

Module Objectives

- To understand the concepts of building blocks
  - Architecture Building Blocks
  - Solution Building Blocks
- To understand their role within the ADM
- A comparison with Architecture Patterns

Building Block Characteristics

- A **package of functionality** defined to meet the business needs across an organization
- A building block has published **interfaces** to access functionality
- A building block may **interoperate** with other, inter-dependent building blocks

A Good Building Block

- Considers **implementation** and **usage** and evolves to exploit technology and standards
- May be **assembled** from or a subassembly of other building blocks
- Is **reusable** and **replaceable**

Building Blocks

- The way in which **functionality**, **products** and **custom developments** are assembled into building blocks varies widely
- Every organization must decide for itself the arrangement
- A good choice can lead to improvements in system **integration**, **interoperability** and **flexibility**

Building Blocks

- **Systems** are built from **collection** of building blocks
- They can be defined at many levels of detail
  - **Groupings** at the functional such as a customer database are known as **Architecture Building Blocks**
  - **Real products** or specific custom developments are known as **Solution Building Blocks**

Architecture Building Blocks (ABBs)

- Architecture **documentation** and **models** from the enterprise's **Architecture Continuum**
- They are defined or selecting during application of ADM
  - Mainly in **Phases A, B, C and D**
- The characteristics are as follows
  - They deliver what **functionality** will be implemented
  - They capture business and technical **requirements**
  - They are **technology-aware**
  - They direct and guide the development **of Solution Building Blocks**

ABB Specifications

- Fundamental **functionality** and **attributes**: semantics, unambiguous, including security capability and manageability
- **Interfaces**: chose set, supplied (APIs, data formats, protocols, hardware interfaces, standards)
- **Dependent building blocks** with required functionality and named interfaces
- Map to business/organization **entities** and po**licies**

Solution Building Blocks (SBBs)

- SBBs relate to the **Solutions Continuum**
- They can either be **produced** or **developed**
- The characteristics are as follows
  - They define what products and components will implement the functionality
  - They define the **implementation**
  - They fulfill business **requirements**
  - They are product or **vendor-aware**

SBB Specifications

- Specific **functionality** and **attributes**
- **Interfaces**: the implemented set
- Required SBBs used with required functionality and names of interfaces used
- Mapping from the SBBs to the **IT topology** and operational policies
- Specifications of attributes shared such as **security**, **manageability**, **scalability**
- Performance, configurability
- Design **drivers** and **constraints** including physical architecture
- Relationships between the **SBBs** and **ABBs**

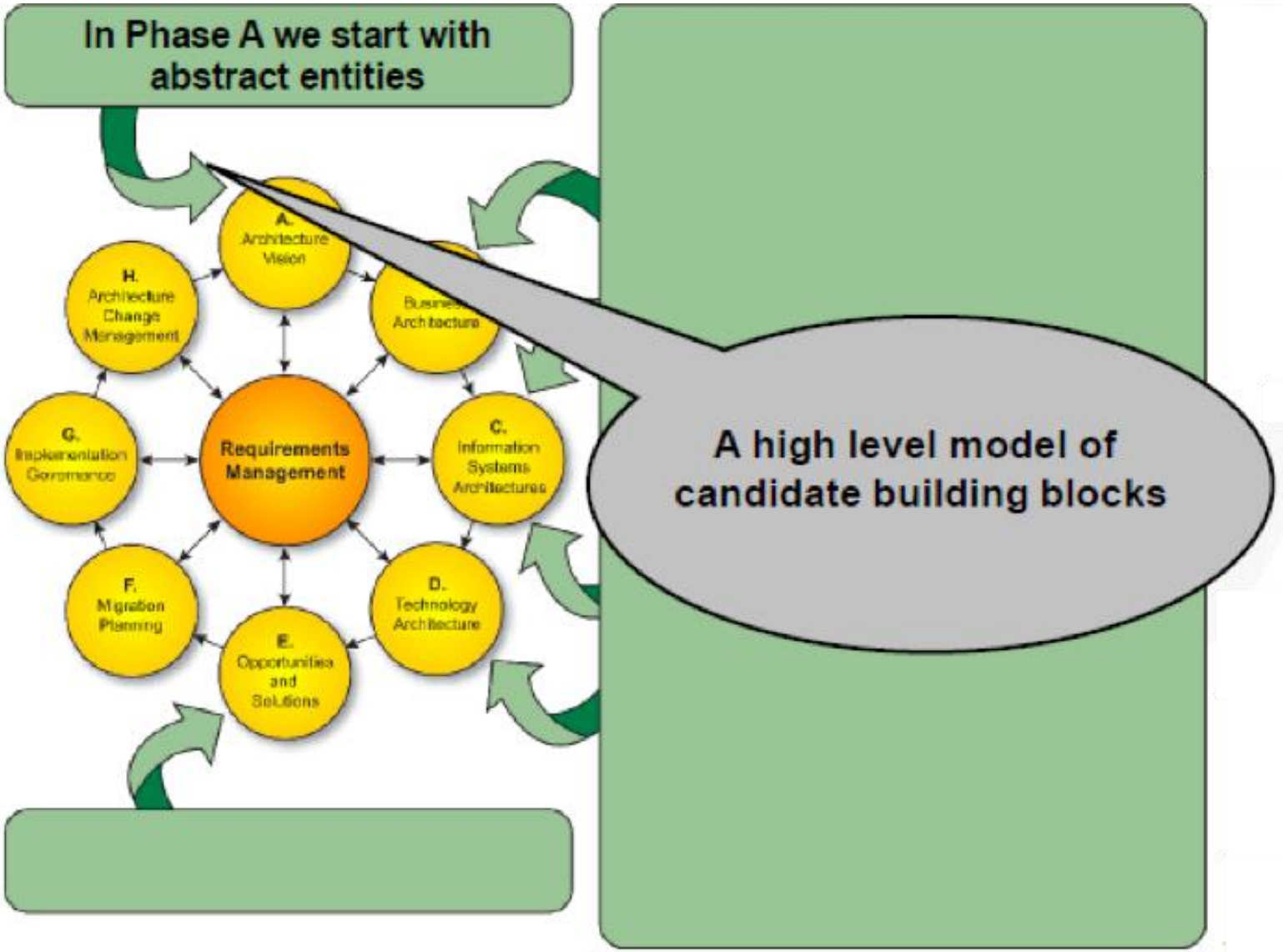
Building Blocks and the ADM

- An architecture is a **set of building blocks**
  - Depicted in an architectural model
  - A specification of how those building blocks are connected to meet the **overall requirements** of an information system
- The various building blocks in an architecture specify the services required in an enterprise specific system
- The following general principles should apply
  - An architecture **need only contain** building blocks to implement those services it requires
  - Building blocks may require one or more parts of a service identified in the architecture
  - Building blocks should **conform to standards**

Building Block Design

- The process of identifying building blocks includes looking for **collections of functions** which require **integration**
- Consider there classes of building blocks
  - **Re-usable** building blocks such as legacy items
  - Building blocks to be developed (**New**)
  - Building blocks to be purchased (**COTS**)
- Use the **desired level of integration** to decide how to bind functions into building blocks

ADM and Building Blocks



ADM and Building Blocks

In Phase A we start with abstract entities



In Phases B, C and D:

Building blocks within the Business, Data, Applications and Technology Architectures are evolved to a common pattern of steps

Step 1: Select Reference Models, Viewpoints and Tools

TOGAF provides example catalogs, matrices and diagrams for use

ADM and Building Blocks



Step 2: Develop Baseline Architecture Description

Develop a high-level model of existing building blocks, re-using definitions from the Architecture Repository where they are available

ADM and Building Blocks



Step 3: Develop Target Architecture Description

Develop view of required building blocks through the creation of catalogs, matrices and diagrams

Fully document each building block

Document rationale for building block decisions

Identify the impacted building blocks within the Architecture Repository, re-using where appropriate

Where necessary, define new building blocks

Select standards for each building block

Document mapping of building blocks to Architecture Landscape

Identify building blocks for re-use, publish as either standards or reference models

ADM and Building Blocks



Step 4: Perform Gap Analysis

Identify building blocks carried over

Identify eliminated building blocks

Identify new building blocks

Identify gaps and determine realization approach

ADM and Building Blocks



Step 5: Define Candidate Roadmap Components

Step 6: Resolve Impacts across the Architecture Landscape

Step 7: Formal Stakeholder Review

Step 8: Finalize the Architecture

Step 9: Create the Architecture Definition Document

ADM and Building Blocks

In Phase A we start with abstract entities



In Phases B, C and D:

Building blocks within the Business, Data, Applications and Technology Architectures are evolved to a common pattern of steps

Building Block architecture in ABB and SBB forms

Associate building blocks with work packages that will address the gaps

Architecture Patterns

- **Pattern**: defined as "**an idea** that has been useful in ne practical context and will probably be useful in others"
- In TOGAF, patterns are considered to be a way of putting building **blocks into context**; for example, to describe a re-usable solution to a problem
- Building blocks are **what you use**: patterns can tell you **how you use them**, when, why, and what trade-offs you have to make in doing so